

OPERATOR'S MANUAL

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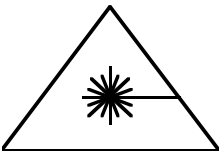
CHAPTER SEVEN SYSTEM SOFTWARE

Introduction

This chapter provides an overview of the Infinity laser system Graphical User Interface (GUI). It is divided into three major sections. The first section contains general interface information, including computer hardware requirements. Section two presents a detailed description of the interface component windows. The chapter concludes with a discussion of the Infinity Command Language. It is advisable to read the first two sections of this chapter before attempting to start the laser for the first time.



The reader is directed to the “readme.txt” file and the “Doc” folder, located on the Infinity CD, for software release notes and installation instructions.



By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than it is specified to provide. Doing this will, however, permanently damage optical components. Do not run the system at higher energies than what the system is rated for.

General Interface Information

The Graphic User Interface (GUI) controlling the Infinity laser system is divided into five main windows. When the program is started only the Infinity Control window is initially displayed. The other windows, Infinity Application Interface, Infinity System Monitor, Infinity Service and Infinity About are accessed by clicking on their respective buttons, “App...”, “Monitor...”, “Service...”, and “About...” in the main control window. The organization of these windows is shown in Figure 7-1.

Even though all of these windows can be open at the same time only one can be active at a time. The information displayed is continuously updated in both the active and the inactive windows. A window is made active by positioning the mouse cursor anywhere on the window and depressing the left mouse button. In this manual the term “clicking” is used to refer to this maneuver. Clicking on a button such as “Off” while the window in which it is located is inactive will not only activate the window, it will also perform the action attached to the button. Just as with any Windows program the Infinity windows can be moved on the desktop and resized. This is particularly useful when multiple windows are open.

Figure 7-2 shows general controls available in every Windows window. These are used to resize, move, open and close windows. They are accessed by clicking on the symbol in the upper left corner. A window can also be minimized, maximized, and closed by clicking on the icons in the upper left corner of each window.

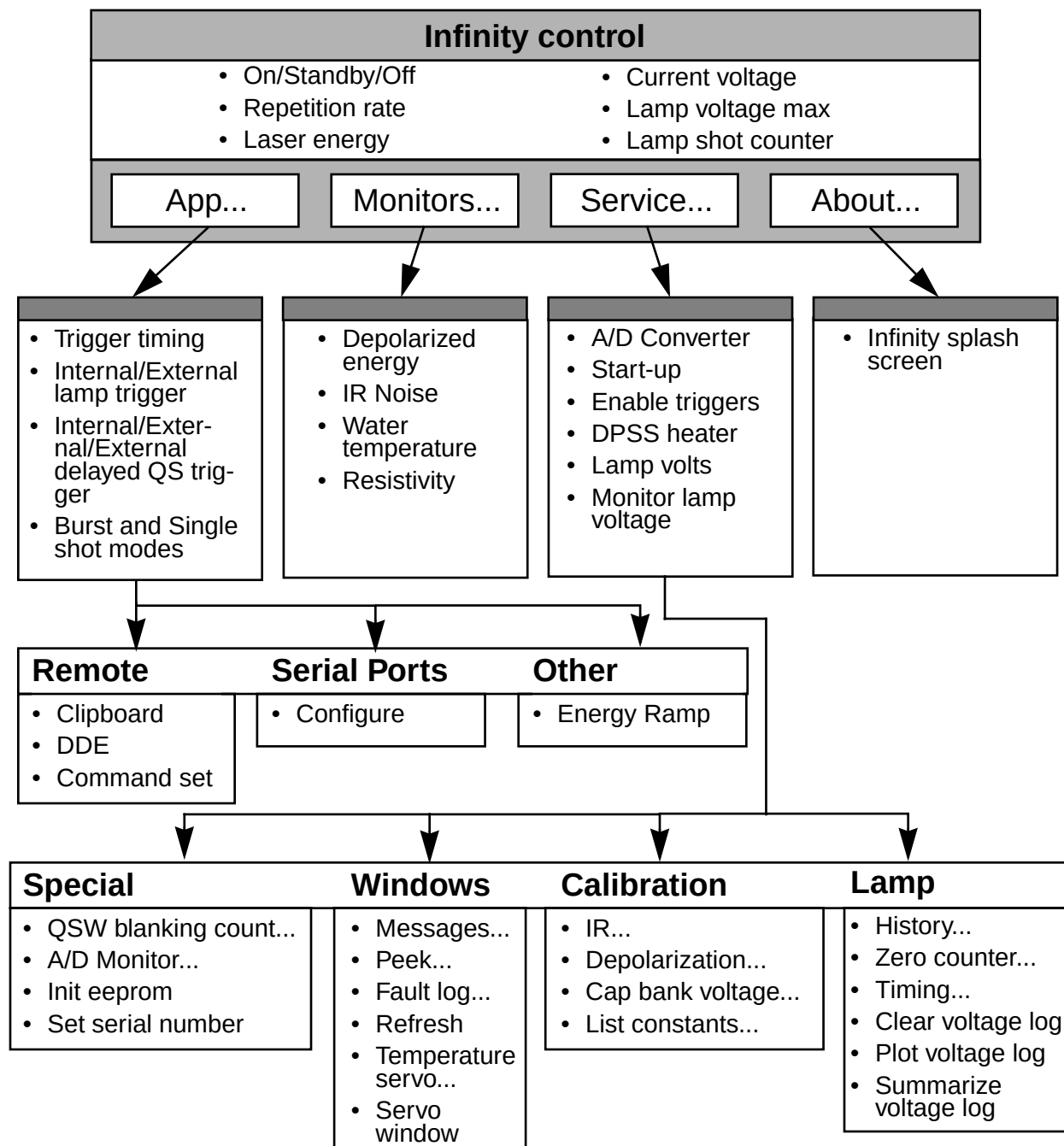


Figure 7-1. Infinity GUI Window Hierarchy.

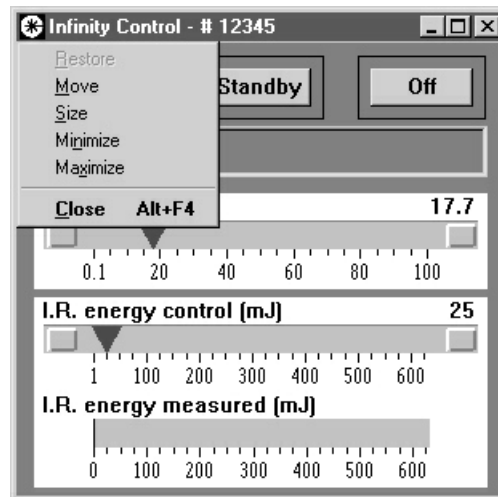


Figure 7-2. General Window Controls.

Menu Bars

Some of the Infinity windows have a menu bar, see for example Figure 7-3. The “drop down” menus are activated clicking on them or by depressing the “Alt” key on the keyboard and simultaneously depressing the underlined letter. Once a drop down menu is activated the desired function can then be chosen in the same manner. Depressing the “Esc” key will deactivate the window menu.

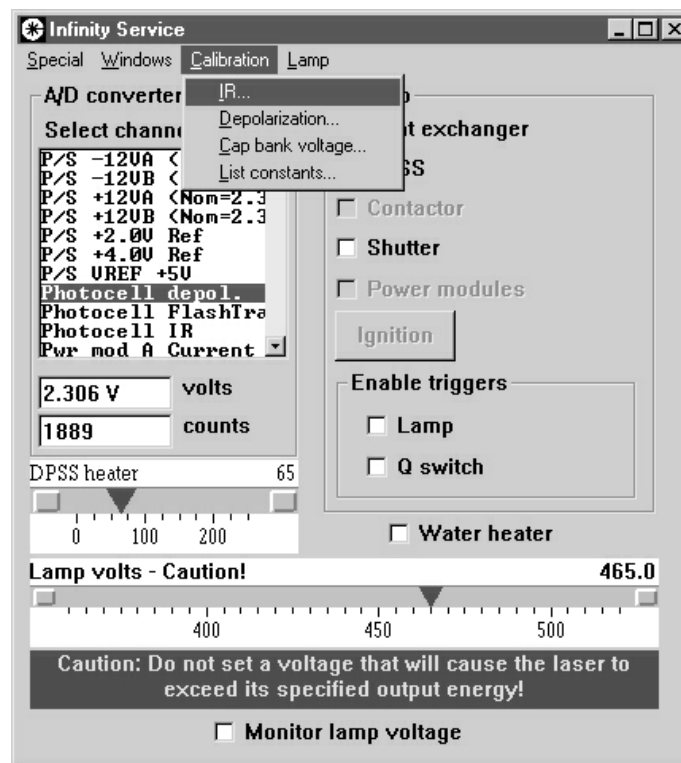


Figure 7-3. Menu Available under Calibration in the Service Window.

Slide Bars

Most parameters in the Infinity windows are controlled through the use of slide bars such as the one shown in Figure 7-4. This particular panel is taken from the main control window and it is used to set the repetition rate. It can be done in three different ways. The numbers given below are only valid for this particular panel. In other panels the parameter value will change by different amounts when the same operation is carried out. The relationship between the magnitude of the changes is roughly the same.

1. Positioning the mouse cursor on the triangular indicator, item 1 in Figure 7-4, and depressing the mouse button allows the indicator to be dragged along the slide bar with a resolution of 1 Hz.
2. Positioning the mouse cursor on the area to the left or right of the indicator, item 2 in Figure 7-4, and clicking on the mouse button will increase or decrease the repetition rate by 10 Hz. If the repetition rate is set to a non-integer number it will change to the nearest even multiple of 10. Holding the mouse button down will move the repetition rate rapidly in steps of 10 Hz.
3. Positioning the cursor on the left or right box, item 3 in Figure 7-4, at the end of the scale and clicking on the mouse button will increase or decrease the repetition rate by 0.1 Hz. Holding the mouse button down will move the repetition rate rapidly in steps of 0.1 Hz.

The number set is displayed in the upper right corner.

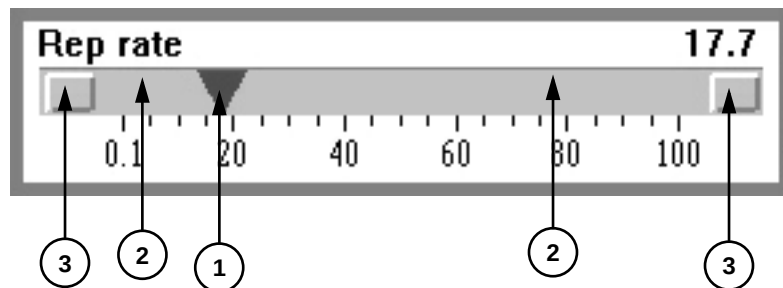


Figure 7-4. Active Areas in Infinity Slide Bar.

Computer Hardware Requirements

The Infinity software is designed to run under the Microsoft® Windows 95 and NT operating systems on an IBM compatible computer. In order to install and run the Infinity software the computer system should meet the following hardware requirements:

- Minimum of a 486/DX4 processor. Although Windows will run on such a computer, a Pentium processor is strongly recommended.
- A minimum of 16 megabytes (MB) of RAM.
- A minimum of 30 MB free hard disk space.
- A CD ROM drive.
- A Microsoft® compatible mouse or trackball. The program uses a Graphical User Interface (GUI) which can only be controlled by the use of a mouse.

Program Windows

The following section presents each of the windows, as well as the functionality of each of the window controls, in the Infinity GUI.

Main Control Window

This window is automatically displayed when the program is run. It is used for switching on and off the laser, for setting the repetition rate and output energy, and for accessing all other windows. It also contains a status display and two shot counters for the flash lamp, one which can be reset and one which is accumulative. The content of this window is described in Table 7-1. Termination of the program is achieved by either pressing Alt and F4 at the same time or clicking on the symbol in the outer right corner. Note that the laser system serial number is displayed in the title bar.

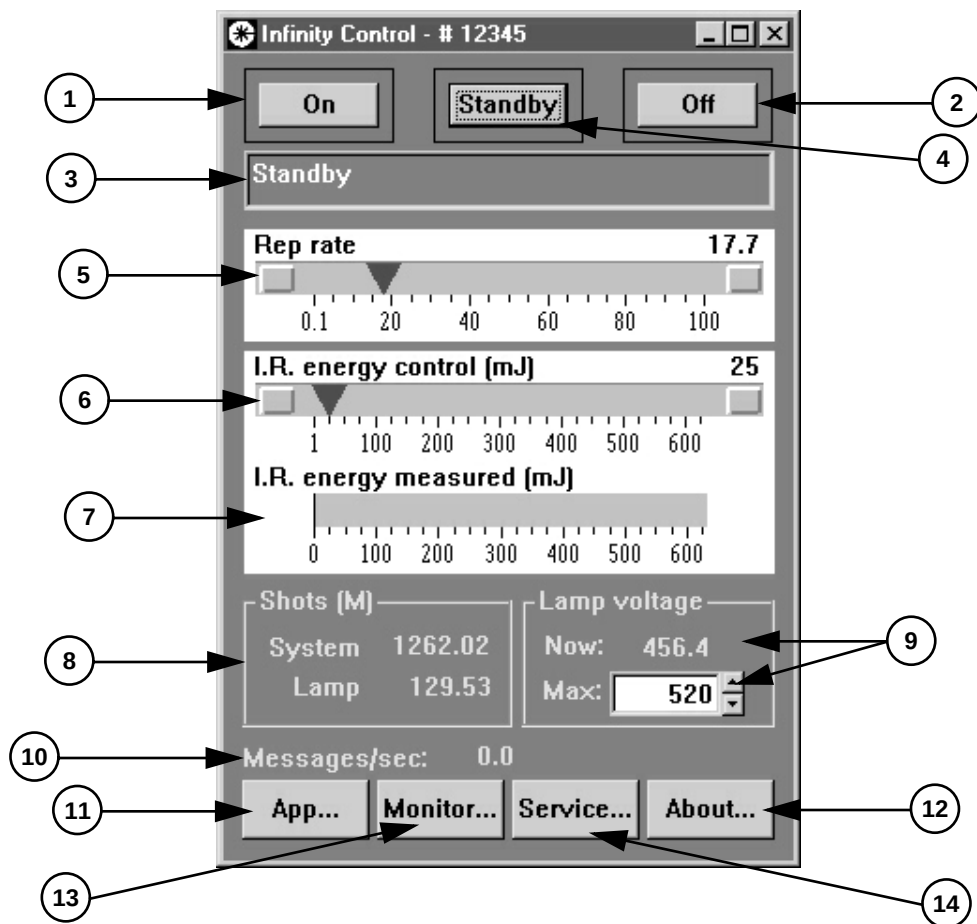


Figure 7-5. Infinity Main Control Window.

Table 7-1. Functions in the Infinity Main Control Window.

#	CONTROL	FUNCTION
1	On	Pressing the On button initiates a chain of events that after a ~5 sec. delay starts the laser. The laser will lase at the repetition rate and power level set on the slide bars in panels 5 and 6, respectively. The On button can be pressed directly if the laser is turned off; it is not necessary to go to standby mode first.
2	Off	Pressing the Off button will shut down all sub-systems. Lasing is terminated immediately.
3	Display	System status messages are displayed in this window.
4	Standby	Pressing the Standby command will turn on the water pump, master oscillator (DPSS), and simmer the flash lamp (assuming that the laser is Off). In Standby mode all systems are active except that the flash lamp and Q-switch triggers are disabled.
5	Repetition rate panel	The repetition rate of the laser is set in this panel. It can be done in three different ways. 1. Positioning the mouse cursor on the triangular indicator and depressing the mouse button allows the indicator to be moved along the slide bar with a resolution of 1 Hz. 2. Positioning the mouse cursor on the area to the left or right of the indicator and clicking on the mouse button will increase or decrease the repetition rate by 10 Hz. An exception to this occurs if the repetition rate is a non-integer multiple of 10, e.g. 83 Hz or 20.1 Hz. Clicking in either area will then change the repetition rate to the nearest higher multiple of 10, i.e. 83 Hz changes to 80 Hz or 90 Hz and 20.1 Hz changes to 20 Hz or 40 Hz. 3. Positioning the cursor on the left or right box at the end of the scale and clicking on the mouse button will increase or decrease the repetition rate by 0.1 Hz. When external lamp triggering is chosen the color of the slide bar changes from white to grey.
6	Pulse energy panel	In this panel the pulse energy of the laser is set. It can be done in three different ways. 1. Positioning the mouse cursor on the triangular indicator and depressing the mouse button allows the indicator to be moved along the bar with a resolution of 5 mJ. 2. Positioning the mouse cursor on the area to the left or right of the indicator and clicking on the mouse button will increase or decrease the pulse energy by 25 mJ. 3. Positioning the cursor on the left or right arrow at the end of the scale and clicking on the mouse button will increase or decrease the pulse energy by 1 mJ. The number set is displayed in the upper right corner. Setting the output energy in this panel automatically enables the LightLoop servo. If the lamp voltage is set manually in the service window Light Loop is deactivated, the pulse energy is set to 0 mJ in panel 6 and the color of the slide bar changes from white to grey.

#	CONTROL	FUNCTION
7	Energy meter	Displays the infrared energy as measured by the built-in energy meter. It can be different than the set energy, displayed in panel 6. A calibration window is available in the menu bar in the Service window, see Chapter Four for IR calibration details.
8	Lamp shot counter	System. Number of shots the system has fired measured in 10E6. This counter can not be changed by the user and is unaffected by any changes in the system. Lamp. Number of shots the flash lamp has fired measured in 10E6. Reset the counter each time the flash lamp is changed.
9	Lamp voltage	Now: The actual voltage applied to the flash lamp. It is set either manually in the Service window or by the on-board microprocessor. In the latter case the processor sets the voltage to what is necessary in order to produce the output chosen in panel 6. In neither case can the voltage be set higher than the lamp max voltage, shown in the same panel. When the difference between actual and max voltage is less than 5 V the background color changes to red. Max: This voltage represents the maximum that can be applied to the flash lamp. Clicking on the up or down arrow will increase or decrease it by 5 V. Minimum setting is 500 V and maximum 570 V.
10	Messages / sec:	The average number of messages transferred over the serial cable between the power supply and the computer.
11	App...	Opens a window in which 1. the timing of two independently variable synchronization signals can be set (the pre-trigger and the high precision trigger). 2. external or internal triggering of the flash lamp and the Q-switch is chosen. See Table 7-3. 3. the delay time for internal delayed trigger mode is set. See Table 7-3. 4. the burst and single shot modes of operation can be selected. 5. the COM ports and communication with another computer are configured.
12	About...	Displays the Infinity "Splash". Gives the software version number and copy right information.
13	Monitor...	Opens a window containing bar graphs displaying depolarized energy, pulse energy noise, water temperature, and water resistivity.
14	Service...	Opens the Infinity Service window. Sub-systems can be switched on and off separately, the lamp voltage and the DPSS temperature can be set manually and various test point values read out.

Timing, Triggering, Burst and Single Shot modes

This window is accessed by pressing the “App...” button in the main control window. The controls in it allow you to:

- Choose between internal and external triggering of the flash lamp and the Q-switch
- Set the timing of the two outgoing synchronization pulses, available on the power supply
- Set the internal delay when using the laser in “External delay” mode
- Set the system to provide a single shot or “n” number of shots on demand
- Configure the external communication settings of the system

Resizing this window also resizes the timer panels. To set the timers with optimum precision enlarge the window by clicking on the center window symbol in the upper right hand corner. See Chapter Four, Daily Operation, for details about timing and triggering.

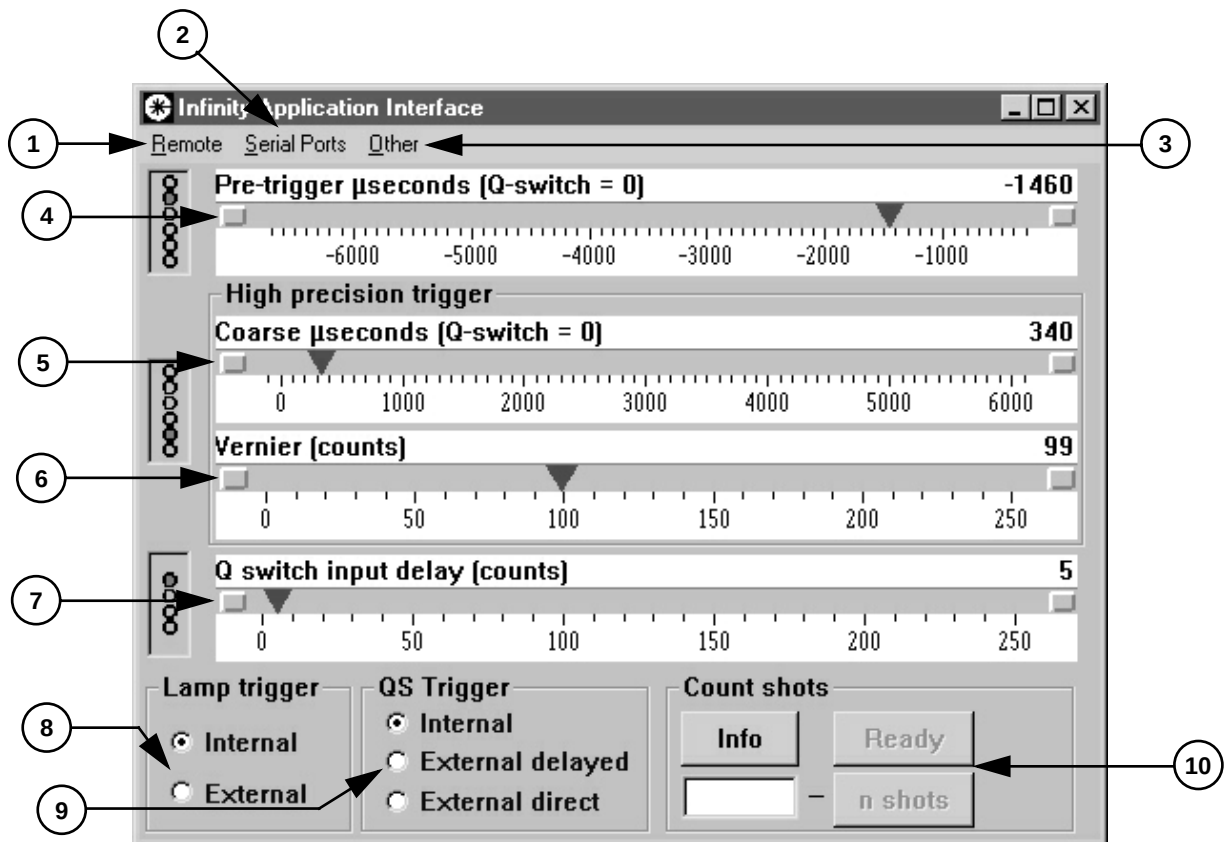


Figure 7-6. Timing and Triggering Window.

Table 7-2. Functions in the Timing and Triggering Window.

#	CONTROL	FUNCTION
1	Remote	Clipboard Sends the content of a pasted clipboard as commands to the laser. The clipboard can be created in any word processor or text editor. All commands are processed sequentially. DDE (Dynamic Data Exchange) Toggles between DDE on and off. In the On mode the Infinity system responds to DDE commands. Command set Opens a new window in which all commands available for remote control of the laser are listed.
2	Serial ports	Configure Opens a new window in which • The COM port that the laser is controlled from can be chosen • The communication protocol and which COM port the remote is connected to can be set. See Figure 7-7 and Table 7-3.
3	Other	Energy Ramp Opens a new window in which the Infinity laser system can be programmed to ramp its IR energy. The user can set the start and stop energies, as well as the time of the total ramp and the dwell time at the final energy. See Figure 7-8 and Table 7-4.
4	Pre-trigger	Programmable outgoing trigger referenced to the Q-switch trigger. It can precede the Q-switch trigger by between 6.77 ms and ~240 μs. Available at the LAMP TRIG SYNCH BNC on the power supply.
5	High precision trigger	Programmable outgoing trigger referenced to the Q-switch trigger. It can precede the laser pulse by a maximum of 150 μs and supercede it by a maximum of 6.15 ms. Available at the VAR QS SYNCH BNC on the power supply.
6	Vernier	High-resolution control of the outgoing Q-switch sync trigger. One count corresponds to ~0.5 ns. The total range is ~100 ns.
7	Q-switch input delay	This control allows the user to delay the externally supplied Q-switch sync pulse before it is allowed to trigger the Q-switch. One count corresponds to ~0.1 ns. The total range is ~25 ns.
8	Lamp Trigger	These buttons toggle between Internal and External triggering of the flash lamp. Internal. The laser provides the flash lamp trigger pulse. External. The laser waits for a user-supplied TTL trigger to the Lamp Trig BNC on the power supply. Triggering occurs on the falling edge of the pulse.

#	CONTROL	FUNCTION
9	QS Trigger	This button toggles between Internal, External and External delayed triggering of the Q-switch. Internal. The laser provides the Q-switch trigger. The timing of it is set in the Lamp timing window available from the Service window. External. The laser waits for a user-supplied TTL trigger to the QS Trig BNC on the power supply. Triggering occurs on the falling edge of the pulse. External delayed. Offers the possibility of delaying the external trigger before it triggers the Q-switch. The internal delay time is set using the slide bar (7) described above.
10	Count shots	These buttons allow you to fire a singleshoot or to program the laser system to provide “n” number of shots. The labels presented on these buttons will depend on the EPROM loaded into the power supply. If a v1.022 EPROM is not used, the buttons will appear as indicated in Figure 7-6 and the software option will not be implemented. If a v1.022 EPROM is used, the buttons will be labeled as indicated below (arrow indicates label change for proper EPROM): Ready or Standby: Pressing of this button places the laser system in “armed” mode. Info → 1 shot: Pressing this button causes the laser to fire 1 shot at the pulse energy set in the main control window. n shots. Pressing this button causes the laser to fire “n” shots, set in the input box below the 1 shot button, at the pulse energy and repetition rate set in the main control window.

Serial Port Configuration

The controls in this window allow the user to configure the external communication parameters of the laser system. See the end of this chapter for additional information on remote controlling the Infinity laser system.

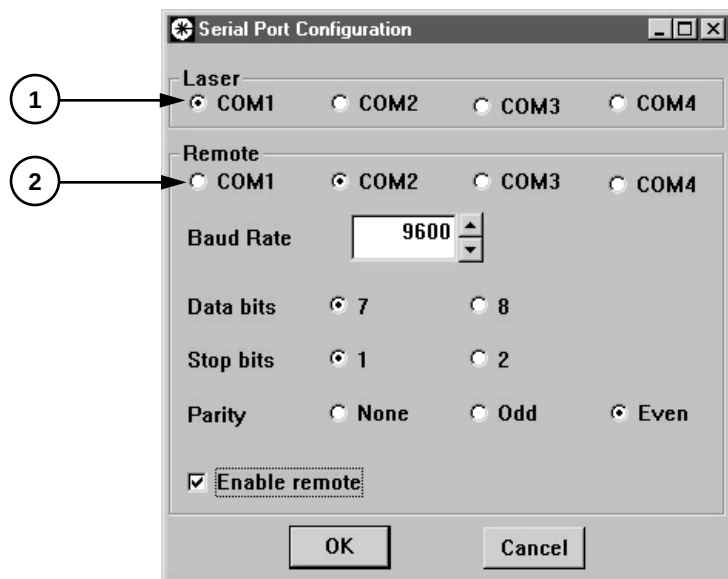


Figure 7-7. Serial Port Configuration Window.

Table 7-3. Functions in the Serial Ports Window.

#	CONTROL	FUNCTION
1	Laser	COM Selects the COM port to which the power supply is connected.
2	Remote	COM Selects the COM port to which the remote computer is connected. Baud Rate Selects the Baud rate used for communication between the remote and the PC. Data bits Selects the number of data bits used for communication between the remote and the PC. Stop bits Selects the number of stop bits used for communication between the remote and the PC. Parity Selects the parity used for communication between the remote and the PC. Enable remote When this box is checked the Infinity PC is controlled from the remote PC.

Energy Ramp

The controls in this window allow the user to configure the Infinity so that it ramps its IR energy.

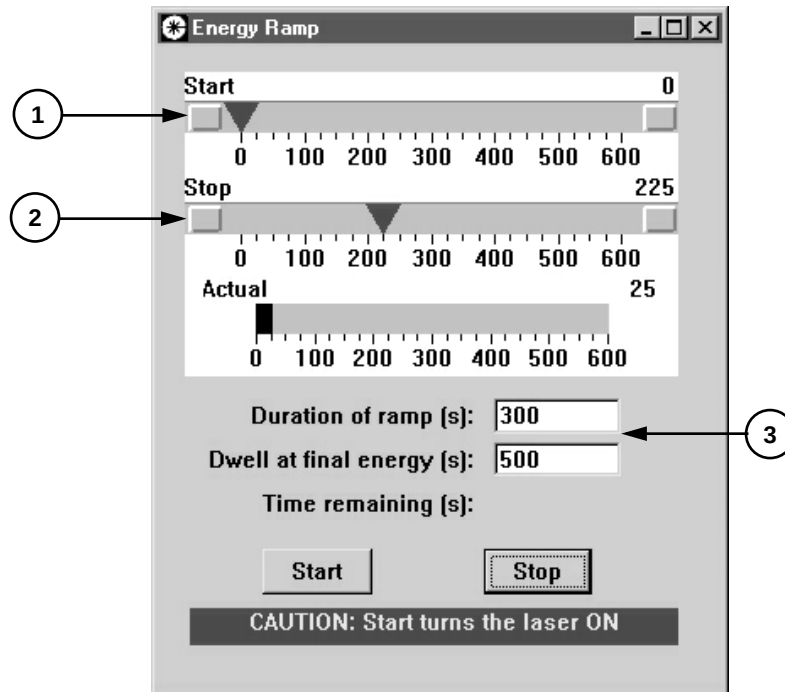


Figure 7-8. Energy Ramp Window.

Table 7-4. Functions in the Energy Ramp Window.

#	CONTROL	FUNCTION
1	Start	Allows the user to select the start energy for the ramp.
2	Stop	Allows the user to set the end energy for the ramp.
3	Duration and Dwell Times	Allows the user to set the total time, in seconds, for the energy ramp as well as the time spent at the “stop” energy before the laser is switched to Standby.

Monitor

This window is accessed by pressing the “Monitor...” button in the main control window. It contains four bar graphs, shown in Figure 7-9, displaying depolarized energy, system energy IR noise, water temperature and water resistivity.

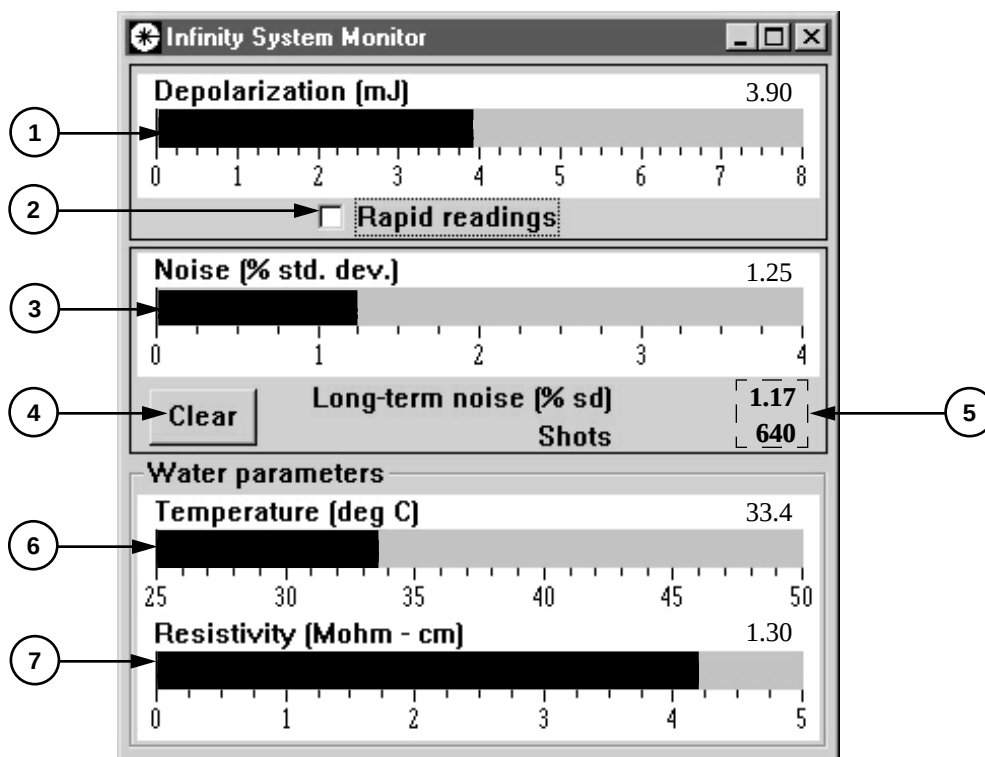


Figure 7-9. Infinity Monitor Window.

Table 7-5. Functions in the Infinity Monitor Window.

#	CONTROL	FUNCTION
1	Depolarization	The laser radiation that after amplification, instead of being reflected of the thin-film polarizer, is transmitted through it and directed back into the Faraday isolator is called depolarized energy. A calibrated photodiode detects this energy and the result is displayed in panel 1. If the depolarized energy is above 6 mJ / pulse see Procedure 4 in Chapter Four, Daily Operation. See Procedure 11 for the calibration procedure.
2	Rapid readings	Activating rapid readings stops the transfer of noise data and uses the freed capacity to speed up the transfer rate of depolarized energy data. This mode of operation is intended to be used when the laser controls are adjusted to minimize the depolarized energy.
3	Noise	The measured standard deviation of the shot-to-shot fluctuations of the IR energy is displayed in panel 3. The noise is only specified for the system when it is operated at maximum energy. If the system is found to be noisier than specifications see Procedure 3 in Chapter Four, Daily Operation.
4	Clear	Clears the long term noise data and restarts the measurement.
5	Long term noise	Shot-to-shot intensity noise calculated from the number of shots shown. After the shot counter has reached 640 the noise data represents a moving average of the most recent 640 shots.
6	Temperature	Temperature of the water in the internal cooling loop. It is measured by a thermocouple located in the reservoir. If the temperature rises above 45°C the laser is turned off.
7	Resistivity	Resistivity of the water in the internal cooling loop. The resistivity should be higher than 3 M Ω /cm. Replace the filter when the resistivity falls to or below 2 M Ω /cm with the adjustment valve fully open. If the resistivity falls below 1.5 M Ω /cm the laser will not operate.

Service Window

This interface is accessed by pressing the “Service...” button in the main interface. It allows the user to switch on and off different components of the system individually, monitor system test points, set the master oscillator temperature and flash lamp voltage. Setting the lamp voltage manually in this window automatically disables flash track and sets the energy read out in the main control window to zero.

In order to access this window the test file “Service.txt” must be present in the Infinity directory. If this file is not present the message “Special authorization is needed to access the Service Window!” will be displayed. The content of the Service.txt file is unimportant.



By manually increasing the flash lamp voltage, output energies significantly higher than specified for the system can be produced. Doing this will permanently damage optical components. Do not run the system at higher energies than what the system is rated for.

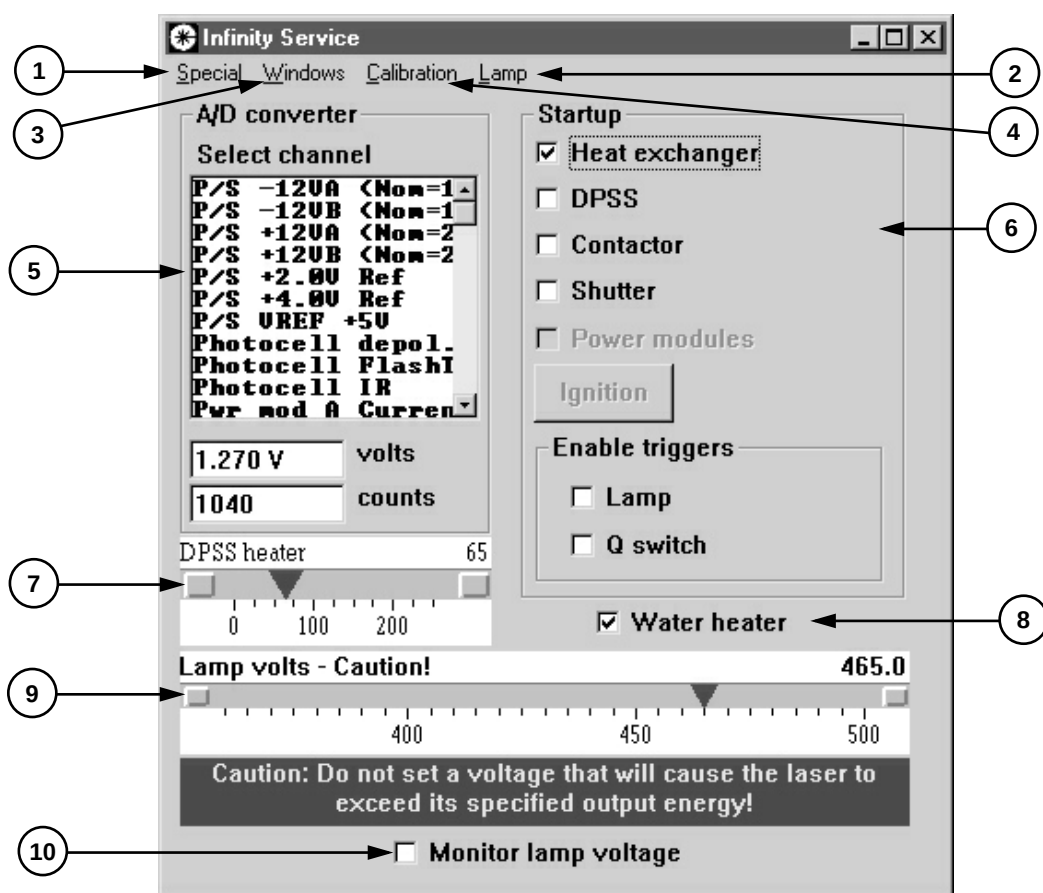


Figure 7-10. Infinity Service Window.

Table 7-6. Functions in the Infinity Service Window.

#	CONTROL	FUNCTION
1	Special	QSW blanking count... Sets the number by which the flash lamp repetition rate will be divided to yield the repetition rate of the laser output. The default value is 1. A / D Monitor... Allows a A / D channel value to be averaged. Parameter is selected from the list in the window. The default cycle value is 50. Init eeprom After a new eeprom has been installed in the power supply this program is run to initiate it. The laser will automatically tell you if it is necessary. Set serial number... Writes the laser systems serial number to the EPROM.
2	Lamp	History... Returns a log of the number of shots fired by each flash lamp used in the laser. Zero counter... Resets the flash lamp shot counter to zero. Reset it after every flash lamp change. Timing... See Figure 7-12 and Table 7-8. Clear voltage log Restarts the collection of lamp voltage data. Plot voltage log Returns a plot of lamp voltage as a function of time. Summarize voltage log Opens a window displaying the lamp voltage change as a function of lamp shots.
3	Windows	Messages... Returns a log of all messages relating to changes in operating parameters sent between the computer and the laser system. Peek... Returns a listing of the status of all digital control lines. Fault log... Opens a window containing a list of all fault messages displayed since the system was installed. Refresh Updates all information in all windows. Temperature servo... Opens a window in which the parameters for the SHG and the THG temperature servos are set. See Figure 7-11 and Table 7-7. Servo Window... Opens the servo window for FlashTrack. See Figure 7-13 and Table 7-9.

#	CONTROL	FUNCTION
4	Calibration	IR... Opens a window in which the built in IR energy meter is calibrated. See Procedure 10 in Chapter Four for the procedure. Depolarization... Opens a window in which the built in depolarized energy meter is calibrated. See Procedure 11 in Chapter Four for the procedure. Cap bank voltage... Opens a window from which the automatic capacitor bank voltage calibration procedure is initiated. It is important that the laser has been on for at least 10 minutes before this procedure is used. List constants... Lists the constants that are used to compute calibrated values for IR energy, depolarized energy and capacitor bank voltage. This window is for reference only. To change the constants, use the calibration procedures 10 and 11 in Chapter Four.
5	A/D Converter	Clicking on a line in the list provides the test point value of the chosen parameter in the display panel below it. The test point is only read once.
6	Startup	The buttons controlling the following systems are automatically enabled or disabled to prevent unsafe operation. It is thus not possible to start the sub-systems in the wrong order. Heat exchanger. Switches on (checked) or off (blank) the pump that circulates the internal cooling water. DPSS. Turns on the laser diode that pumps the master oscillator. Contactor. Engages the main contactor in the power supply. Shutter. Opens the mechanical shutter, located between the thin film polarizer and the spatial filter, in the laser head. No laser light can be emitted when the shutter is closed. Using the shutter allows the user to eliminate lasing without turning off the flash lamp or the oscillator. Going to Standby mode or disable the Q-switch trigger will also temporarily eliminate lasing. Power modules. Turns on the two booster boards that supply the charging voltage to the capacitor bank. Ignition. Simmers the lamp. If the cover is off (requires the cover interlock to be defeated) this can be seen as a steady intense glow from the lamp. Simmering of the lamp means that an ionized channel has been established between the anode and cathode in the lamp. Lamp. Enables or disables the lamp trigger. Q-switch must be off. Q switch. Enables or disables the Q-switch trigger. The lamp must be turned on before the Q-switch in order for the amplifier to reach partial thermal equilibrium before the Q-switched pulse is injected into it.
7	DPSS Heater	Slide bar used to change the temperature of the master oscillator. Optimum operating temperature is set at the factory. It only needs to be changed if problems with mode jumps occur. See Chapter Four, Procedures 3 and 9. The temperature range is 35°C–55°C.

#	CONTROL	FUNCTION
8	Water heater	Turns on the heater in the water reservoir. This is always done automatically by the system if needed.
9	Lamp volts	Allows the flash lamp voltage to be set manually. Lightloop is automatically disabled. It is enabled by setting the output energy in the main control window (mJ). By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than what it is specified to give. Doing this will permanently damage optical components. Do not run the system at higher energies than what the system is rated for.
10	Monitor lamp voltage	Toggles between lamp monitor off and on. When On, lamp voltage as a function of time is monitored and stored in a file. The content of the data file can later be plotted using the plot command under the lamp menu title in the Service... window. This feature is useful for tracking lamp aging. Note that this feature is automatically disabled if the Service window is closed.

Crystal Temperature Window

The crystal mounts temperature servo set point is set in the Crystal temperature window. This window is accessed from under the Windows menu in the Service window. The temperature (DAC counts) is factory set.

If the servo after 15 minutes has not been able to regulate the crystal mount temperature it will assume that there are no mounts present and set the drive signal to zero.

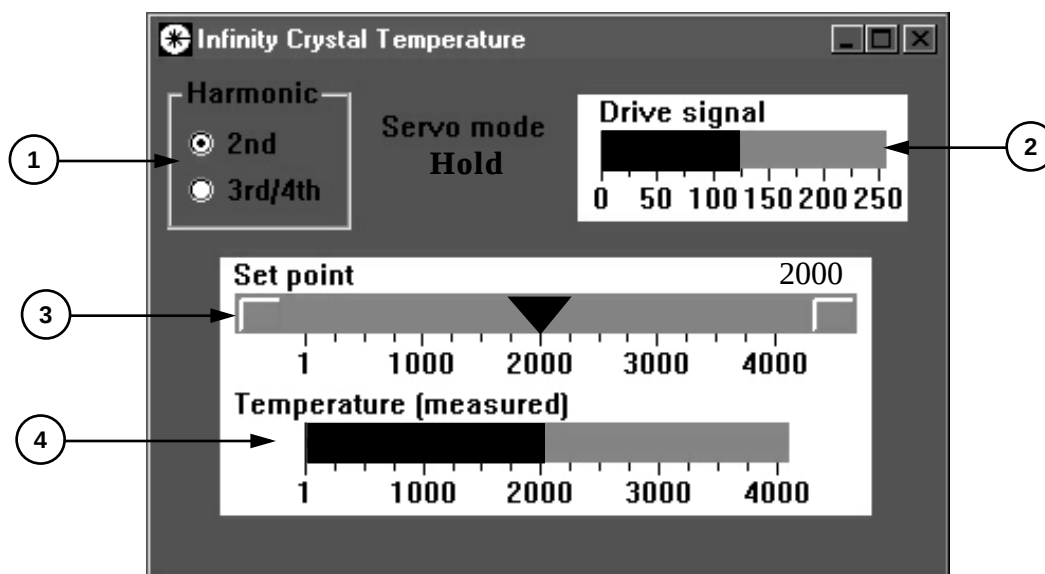


Figure 7-11. Crystal Temperature Window.

Table 7-7. Functions in the Crystal Temperature Window.

#	CONTROL	FUNCTION
1	Harmonic	Selects which mount the window controls addresses.
2	Drive signal	Proportional to the difference between set and actual temperature.
3	Set point	The temperature set point for the crystal mount.
4	Temperature	The actual temperature of the crystal mount. When the servo is working properly the temperature should be equal to the set temperature.

Lamp Timing Window

The controls in the lamp timing window allow the user to customize the flash lamp pulse width and the Q-switch trigger position. They are set at the factory so that the laser performs optimally. Under normal usage of the laser there should be no need to change them. See Chapter Six, Maintenance, for additional information on setting the “System Timing”.



Modifying the parameters contained in the lamp timer window will affect laser performance. Deviation from factory settings may result in lower output power and unstable DPSS laser performance.

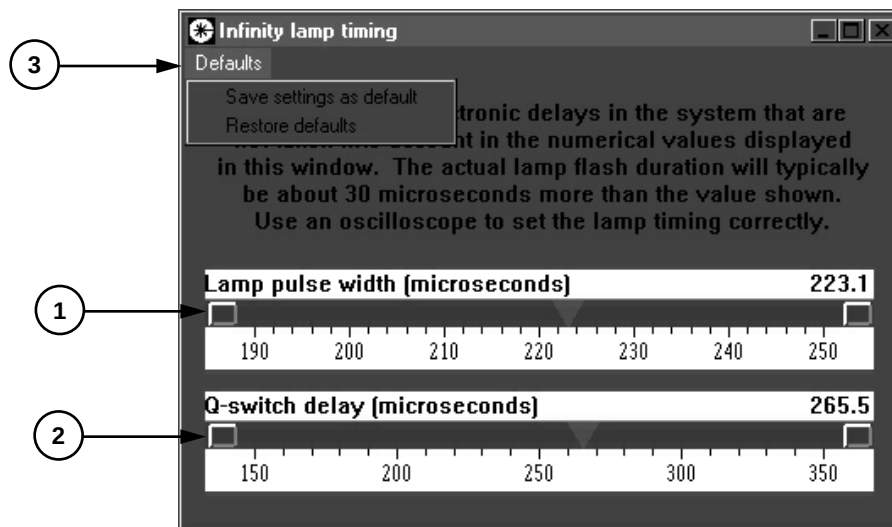


Figure 7-12. Lamp Timing Window.

Table 7-8. Functions in the Lamp Timing Window.

#	CONTROL	FUNCTION
1	Lamp pulse width	Sets the lamp emission time by controlling the duration of the current pulse from the capacitor bank. The width, as measured on an oscilloscope, should be 250 μ s.
2	Q-switch delay	The setting of the Q-switch delay determines when, during the flash lamp emission, the Q-switch is triggered. The optimum setting which is set at the factory, is 4 μ s before the flash lamp is turned off.
3	Defaults menu	Allows the user to reset the timing values to the default values and to save the current values as default values.

FlashTrack Servo Controls

FlashTrack is a control system that reduces shot-to-shot intensity fluctuations. Figure 7-13 gives the control window for this feature. See Chapter Two for additional information on this feature.

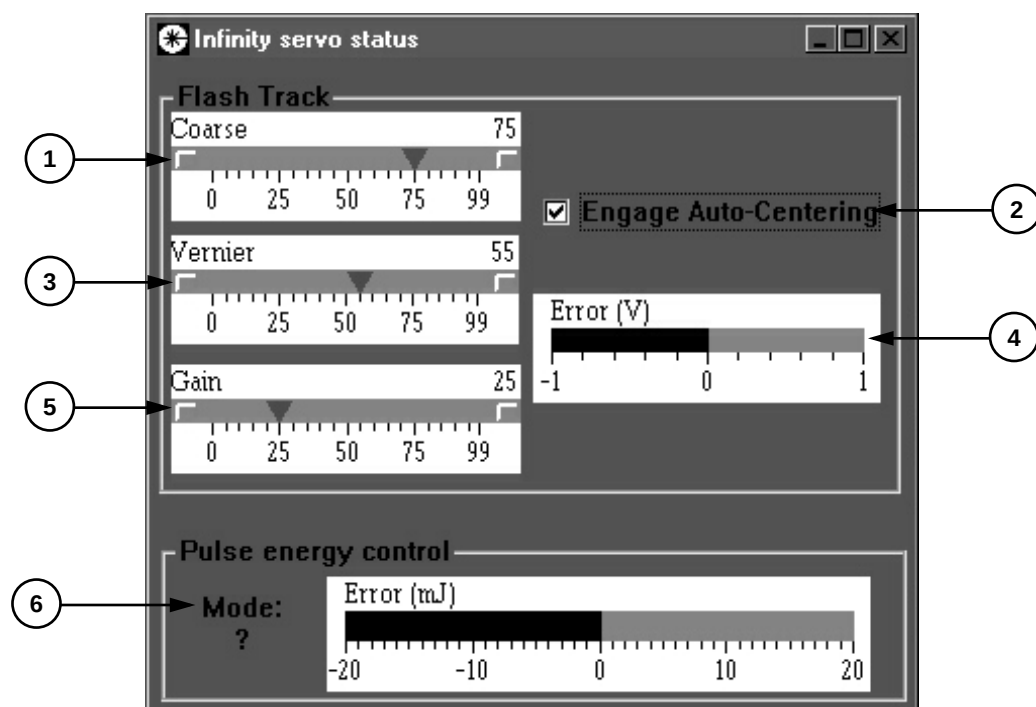


Figure 7-13. Flash Track Window.

Table 7-9. Functions in the Flash Track Servo Window.

#	CONTROL	FUNCTION
1	Coarse	Coarse adjustment for level shift of the photo diode signal. Flash Track can be turned off by setting coarse to zero. It is automatically optimized when auto-centering is engaged.
2	Engage Auto Centering	Automatically optimizes the coarse and vernier settings. Optimum settings yield a photocell Flash Track voltage of close to 2.5 V (center of the A/D converter range). This voltage can also be read out through the A/D converter in the Service window.
3	Vernier	Fine adjustment for level shift of the diode signal. It is automatically optimized when auto-centering is engaged.
4	Error	Voltage proportional to the energy correction applied.
5	Gain	The gain setting determines the amount of energy correction applied for a given error signal. Optimize it by monitoring the intensity noise, panel 3 in Figure 7-9.
6	Pulse energy control	The difference between measured energy and set energy. Not active when the lamp voltage is set manually.

Laser Control Methods

The Infinity control software provides a message language for controlling the laser system without directly using the Graphical User Interface (GUI). There are three ways to use the message language: First, from another Windows application by means of Dynamic Data Exchange (DDE, an operating system feature); second, from an external computer by RS-232; or third, from command scripts pasted to the Windows clipboard. In each case the set of possible messages is the same.

Messages consist of a string of characters from the ASCII character set. When a message is received by the Infinity GUI, it is parsed by a command processor. If the message is found to be syntactically correct it is dispatched to the rest of the application for processing. Most messages will cause a change to the laser state, which will be reflected in the on-screen user interface.



The user interface remains active at all times. It is therefore possible to freely mix mouse input (normal mode of communicating with the GUI) with command processing.

Messages are of three types: command, query, and control. Commands are used to change the laser state by setting parameters such as repetition rate, pulse energy, and timer values. Queries are used to interrogate the laser for status information. A control message is used to terminate a DDE session or list of clipboard commands.

In the following sections the clipboard, RS-232 and DDE methods of communication are discussed.

Infinity Language Commands

Table 7-10 gives a listing of all of the Infinity Language commands. The commands are presented in the first column of Table 7-10. In column two a brief description of the command is given. Note that command arguments are of two types: *float* and *integer*.

Table 7-10. Infinity Command Language.

Command	Description
End	Terminates message exchange. In the case of RS-232, this closes the receiving port. In the case of DDE, this terminates the DDE conversation.
On	Turns the Infinity laser system on.
Off	Turns the Infinity laser system off.
Standby	Places the Infinity laser system in standby mode.
HighPrecCoarse float	Units are microseconds.
HighPrecVernier integer	Units in DAC counts.
Input Delay integer	Sets Q-switch input delay, in nanoseconds.
LampTrigger integer	Sets lamp trigger mode: 0=internal, 1=external.
PreTrigger float	Sets the Applications window Pre-trigger value.
PulseMj float	Sets Infinity IR output energy.
QswTrigger integer	Sets Q-switch trigger mode: 0=internal, 1=external direct, 2=external delayed.
RepRate float	Sets Infinity repetition rate.
qDepol	Returns the depolarized energy in millijoules. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.
qFaultN integer	Returns the test of one active fault from the indexed list. The argument indexes into the list; the number of active faults can be obtained by the query "qFaults". The list is 1-based (i.e., the first fault is # 1). If an invalid argument is sent with this query an empty string will be returned.
qHighPrecCoarse	Queries High precision coarse trigger value, see the Applications window. Value returned is in microseconds.
qHighPrecVernier	Queries High precision trigger vernier value, see the Applications window. Value returned is in DAC counts.

Command	Description
qInputDelay	Queries High precision trigger Q-switch input delay value, see the Applications window. Value returned is in DAC counts.
qLampShots	Units are in millions of shots.
qLampTrigger	Returns current lamp triggering state of the Infinity laser system: 0=Internal, 1=external.
qLampVMargin	Returns the difference between the current lamp voltage and the maximum lamp voltage as set in the main control window of the GUI. Units are in volts.
qLampVMax	Returns the maximum lamp voltage as set in the main control window of the GUI. Units are in volts.
qLTNoise	Returns the long term noise as a percent. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.
qMjActual	Returns the last measured laser power reading. Units are in millijoules.
qNFaults	Returns the number of active faults, defining the valid range of arguments for the query “qFaultN”. If there are no faults active “0” will be returned.
qNoise	Returns the short term noise, based on the last 80 shots, in percent form.
qPreTrigger	Returns the current value for the Q-switch pretrigger. Units are in microseconds.
qPulseMj	Returns the pulse energy set point (NOT the actual laser power which is returned by qMjActual).
qQswTrigger	Returns the current Q-switch trigger mode: 0=internal, 1=external direct, 2=external delayed.
qRepRate	Returns the current repetition rate.
qShots	Returns the total number of shots on the system in millions.
qState	Returns the current state of the laser system in the form of a string which matches the one shown in the display area of the GUI main control window.
qVersion	Returns the software version number shown in the header of the GUI main control window.
qVolts	Returns the actual lamp voltage.

Command	Description
qWaterRes	Returns the water resistivity in megaohm-cm. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.
qWaterT	Returns the water temperature in degrees centigrade. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.

Sending ClipBoard Messages

An easy method for checking command syntax is to send messages to the GUI by means of the Windows clipboard. A command can be placed into the clipboard from any available text editor (the “Notepad” utility found in the Windows Accessories folder works well for this purpose). Once the test editor has been launched type in the command of interest. Then:

1. using the mouse high-light the command
2. select the “Edit” menu item from the text editor main menu. From the resulting popup menu select “Copy”.

To direct the Infinity program to read the contents of the clipboard, in the GUI, open the “Applications” window. From the menu bar select “Remote” and then, from the resulting drop down menu, select “Clipboard”. If the command syntax is correct the GUI will produce an audible beep and then make the requested change in state. If the syntax of the command is wrong a message box will appear on the screen. The heading of this box will read “Remote syntax error:”. The body text of the message box will contain the message “Command X not understood”. Where “X” is the complete command as read from the clipboard.

When using this option it is important to keep in mind that the GUI will not provide any feed back other than the syntax error message box. In other words, no response is generated to queries or to command arguments that are out of range.

Sending DDE Messages

The Infinity program is a fully functional DDE server capable of communicating with other Windows applications. This section describes the protocol used and is intended to support application programming. It assumes a familiarity with Windows DDE concepts and terminology. For further information, see the Microsoft Windows Software

Development Kit (SDK) help or an equivalent reference source.

The Infinity program acts as a DDE Server supporting a single application /topic pair:

Application:'Infinity'

Topic:'Control'

To start the Infinity DDE server, open the Application window and select the DDE item from the Remote pull-down menu. Before commands can be sent to Infinity, the client program must obtain the window handle of the Infinity DDE server. This is accomplished by means of a **WM_DDE_INITIATE** message containing the application /topic pair ('Infinity'/'Control') just given.

Once communication is established, Infinity processes DDE transactions of type **WM_DDE_REQUEST**. This message takes two parameters:

Format: The only supported format type is CF_TEXT.

Item: A string containing the command to be sent to Infinity. The strings are identical to the RS-232 commands as documented in the Infinity User's Manual. A terminating CR/LF is not required.

In response, the Infinity program sends a DDE transaction of type **WM_DDE_DATA** back to the client window. The returned item has a format type of **CF_TEXT**. The content of the responses is the same as for RS-232 commands, see below.

Sending Messages by RS-232

To enable the Infinity program to respond to RS-232 messages you first must configure the serial port on which the Infinity software is to poll for messages, see Figure 7-7 and Table 7-3.

After configuring the serial port, the Infinity program will try to open the port and then begin to poll the port for incoming messages. If there is a problem with the port initialization an error box will appear on the screen. If this occurs the configuration of the serial port will need to be verified (i.e., base address and IRQ conflicts). The information entered in the Serial Port Configuration window is saved when the program is terminated, so it is not necessary to repeat this procedure each time the GUI is started.

RS-232 Message Syntax

Each RS-232 message to the laser consists of a string of printable ASCII characters terminated by a carriage return (ASCII 13). Leading and trailing "white space" characters are

discarded before the message is processed. Replies from the laser will consist of printable ASCII characters followed by a carriage return and a line feed (ASCII 13 + ASCII 10).

Each message begins with the keyword followed by 0 or 1 arguments, see Table 7-10. If an argument is required, use a space character (ASCII 32) between the keyword and the argument. Keywords are case sensitive. Floating point numbers, when required, consist of an optional minus sign, at least one digit (possibly 0) followed by an optional decimal point and an optional string of trailing digits (i.e., [-]n[.][m]). Scientific notation is not supported. Integers are represented by a sequence of digits with no decimal point.

When the program receives a message that begins with an invalid keyword, the words "Syntax error:" (without the quotes) are transmitted back in the reply followed by the entire contents of the invalid message. When the program receives a message that contains a valid keyword, but an invalid argument, the words "Command error:" (without the quotes) are transmitted back in reply followed by the entire contents of the invalid message. This occurs, for instance, if an attempt is made to set a parameter to a value outside of its valid range. For example, "RepRate 135.2" would cause the reply "Command error: RepRate 135.2" because the upper limit of the repetition rate is 100.0.

If the message is received correctly and executed, the word "OK" is transmitted back in reply.



Every message will cause a reply of some kind. It is very important not to transmit a new message until the previous message has been acknowledged by the GUI.

Some messages query the laser for information (these are identified by a keyword that begins with a lower case "q"). In reply to a query, the requested information will be transmitted back in a message-specific format as described in Table 7-10.

Example RS-232 Program

The following is an example program of how to use a second computer to remotely control the Infinity laser system. To obtain the required number of serial ports, a SIIG serial expansion card (model 101812) with a single DB9 port was installed in the Infinity computer. The new Com port was given the base address 03E8 and was assigned interrupt (IRQ) 5, which is normally reserved for a second parallel port.

The 3-serial ports, and the Windows operating system, were configured to allow:

- the remote computer to be on Com port 1
- the Infinity power supply to be on Com port 2
- the mouse to be on Com port 3

The program was written and executed with the QBasic utility program provided with the Windows 3.1 operating system, installed on a Toshiba laptop computer.

When the example program first starts it polls the Infinity software to determine the current system state, repetition rate, IR pulse energy setting, lamp voltage setting, and number of active faults. Following this the function key menu is displayed, F1 through F7. The action initiated by each key is explained in the menu. If input is required from the user a task specific prompt appears below the function key menu.

When the user terminates the program, the laser system is automatically sent to the “Off” state, the repetition rate is set to 10 Hz, and the IR pulse energy is set to 1 mJ. For reasons of safety, it is always advisable to minimize the repetition rate and IR pulse energy before shutting down the laser system.

' Example remote Infinity program

' Coherent Laser Group (1997)

DECLARE SUB Setcomport ()

DECLARE SUB laserstate ()

DECLARE SUB menu ()

COMMON SHARED string1 AS STRING * 80

COMMON SHARED string2 AS STRING * 80

COMMON SHARED flag1%

KEY(1) ON ' Turn on monitoring of function keys

KEY(2) ON

KEY(3) ON

KEY(4) ON

KEY(5) ON

KEY(6) ON

KEY(7) ON

CALL Setcomport ' Initialize com1

```
start:
    CALL laserstate      ' Get current laser state information
    CALL menu           ' Print function key menu to screen
    flag1% = 0
    DO WHILE flag1% = 0
        '   Defining function key F1 as laser to Standby event.
        ON KEY(1) GOSUB lasertostandby
        '   Defining function key F2 as laser to On event.
        ON KEY(2) GOSUB turnonlaser
        '   Defining function key F3 as laser Off event.
        ON KEY(3) GOSUB turnofflaser
        '   Defining function key F4 as set repetition rate event.
        ON KEY(4) GOSUB setretrate
        '   Defining function key F5 as set pulse energy event.
        ON KEY(5) GOSUB setpulseenergy
        '   Defining function key F6 for screen refresh.
        ON KEY(6) GOSUB redoscreen
        '   Defining function key F7 for program termination.
        ON KEY(7) GOSUB exitroutine
    LOOP
END

                                ' Main menu functions

lasertostandby:
    string1 = "Standby"
    PRINT #1, string1
    string1 = ""
    INPUT #1, string1
RETURN start

turnonlaser:
    string1 = "On"
    PRINT #1, string1
    string1 = ""
    INPUT #1, string1
RETURN start
```


turnofflaser:

```
string1 = "Off"
PRINT #1, string1
string1 = ""
INPUT #1, string1
```

RETURN start

setrepreate:

```
string2 = ""
INPUT "Enter new repetition rate: ", string2
string1 = ""
string1 = "RepRate " + string2
PRINT #1, string1
string1 = ""
INPUT #1, string1
```

RETURN start

setpulseenergy:

```
string2 = ""
INPUT "Enter new IR pulse energy: ", string2
string1 = ""
string1 = "PulseMj " + string2
PRINT #1, string1
string1 = ""
INPUT #1, string1
```

RETURN start

redoscreen: RETURN start

exitroutine:

' Shut off laser

```
string1 = ""
string1 = "Off"
PRINT #1, string1
INPUT #1, string1
```

' Set IR energy to minimum value

```
string1 = ""
string1 = "PulseMj 1"
PRINT #1, string1
INPUT #1, string1
```

```
' Set repetition rate to 10 Hz

string1 = ""
string1 = "RepRate 10"
PRINT #1, string1
INPUT #1, string1

' Disable Com port and set termination flag

COM(1) OFF
flag1% = 1
RETURN

' Program subroutines

SUB laserstate
  CLS 0
  PRINT ""
  PRINT "  Sample RS-232 remote control program for the Infinity laser
system."
  PRINT ""

  ' Print last Infinity system message

  PRINT "Last Infinity system message was: " + string1

  ' Get current laser state

  string1 = ""
  string1 = "qState"
  PRINT #1, string1
  INPUT #1, string2
  string1 = ""
  string1 = "Current state of the laser system: " + string2
  PRINT string1

  ' Get current repetition rate

  string1 = ""
  string1 = "qRepRate"
  PRINT #1, string1
  INPUT #1, string2
  string1 = ""
  string1 = "Current Rep Rate is (Hz): " + string2
  PRINT string1
```

```

                                ' Get IR pulse energy
string1 = ""
string1 = "qPulseMj"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current IR pulse energy is (mJ): " + string2
PRINT string1

                                ' Get current lamp voltage
string1 = ""
string1 = "qVolts"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current lamp voltage setting is (V): " + string2
PRINT string1

                                ' Get current number of active faults
string1 = ""
string1 = "qNFaults"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current number of active faults is: " + string2
PRINT string1
PRINT ""
END SUB

```

SUB menu

```
PRINT "           Active Function Keys"
PRINT ""
PRINT "           F1 : Turn Laser to Standby"
PRINT "           F2 : Turn Laser On"
PRINT "           F3 : Turn Laser Off"
PRINT "           F4 : Set Repetition Rate"
PRINT "           F5 : Set Pulse Energy"
PRINT "           F6 : Refresh Screen"
PRINT "           F7 : Exit Routine"
PRINT ""
PRINT "Make a Function Key Selection to Begin"
```

END SUB

SUB Setcomport

```
OPEN "COM1:9600,E,7,1,CD0,CS0,DS0,OP0,RS,TB2048,RB2048" FOR
RANDOM AS #1
COM(1) ON
END SUB
```